Light weight Email Security Tool using Sandbox

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Final Approval

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Declaration

We hereby declare that this document "**Light weight E-mail security tool using sandbox**" neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers especially our supervisor **Mr. Awais Nawaz** If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

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Dedication

We dedicate this work to our families, especially our parents who have always been these for us, giving us love, support, and encouragement. Our friends have always been there for us, understanding us and motivating us. Our mentors have always inspired us with their knowledge, insight, and dedication. This work is for everyone who seeks knowledge with a purpose, values new ideas, and works hard to be the best. We hope it helps make the world a safer, more responsible, and more informed place in its own way.

Acknowledgement

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Abstract

Email is a widely used communication tool but remains a major target for threats such as phishing, malware, spoofing, and data leakage. While platforms like Gmail and Outlook offer built-in security, they are often limited for learning, customization, or small-scale use, and many existing systems rely on basic signature-based detection that struggles with evolving attacks. This project proposes a lightweight email security tool with an integrated sandbox to analyze suspicious attachments in an isolated environment, detect phishing using regex, heuristics, and header analysis, and verify domain reputation to prevent spoofing. It enhances security through AES-256 encryption for confidentiality and RSA-2048 digital signatures for integrity and authenticity, along with features like adaptive encryption for sensitive data, self-destructing emails, and a user-friendly CLI/dashboard for alerts, logging, and risk scoring. The system is designed to be efficient, modular, and easy to deploy, making it suitable for educational purposes and small organizations with limited resources. Overall, the project aims to bridge the gap between basic protections and complex enterprise solutions by providing an accessible yet powerful prototype for secure email communication, with future enhancements including machine learning-based threat detection, cloud scalability, and multi-platform integration.

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Chapter 1:

Introduction

Chapter 1:

Introduction

Email is a common way to talk to people, but it can be dangerous because of phishing, malware, and spoofing. This project proposes a straightforward email security tool that employs a sandbox to detect threats, examine attachments, and ensure secure communication.

1.1 Opportunity & Stakeholders

Hackers still use email as one of the most common ways to get in. Verizon DBIR 2025 and other reports show that email is still the starting point for a lot of malware and social engineering attacks. Large companies often rely on expensive, high-tech tools such as Proofpoint and Mimecast. However, these solutions are not practical for individuals, students, freelancers, or small businesses that lack significant budgets or technical expertise.
This gap lets us create a simple, lightweight email security tool that focusses on basic threat filtering, rule-based phishing detection, and safe handling of attachments. The goal is not to replace enterprise-level systems, but to add a layer of security that is simple for small-scale users to learn and use.

Stakeholders

Primary Stakeholders:

- Individual users
- Freelancers
- People who own small businesses

Secondary Stakeholders:

- School & Colleges
- NGOs
- Remote workers

Tertiary Stakeholders:

- Students and researchers in cybersecurity
- Open-source contributors

1.2 Motivation and Challenges

Motivations:

- Traditional filters that use signatures don't work against zero-day malware or advanced phishing.
- There is no lightweight tool that combines sandboxing, encryption, and real-time alerts.
- Need for a solution that is cheap and easy to set up

Challenges:

- Finding new phishing methods and harmful attachments.
- Ensuring the scanning process is efficient and does not delay email communication.
- Maintaining confidentiality and integrity of emails while performing security checks.
- Providing an intuitive interface for users to monitor email security and take necessary actions.

1.3 Significance of the Study

The project shows how a security tool with multiple layers can help people and small businesses keep their email safer. The system provides security comparable to enterprise solutions, although in a more streamlined and user-friendly format. It does this by using sandbox-based attachment analysis, phishing detection, header verification, and strong encryption all at the same time. This method lets regular people use advanced email security.

1.4 Goals and Objectives

Goals:

- Make an email security tool that is light enough for individuals and small businesses to use.
- Give basic protection against phishing, malware, and attachments that aren't safe.
- Use alerts, logging, and simple reporting to make users more aware.
- Offer a practical, resource-efficient solution for educational and small-scale use cases.

Objectives:

- Develop a modular architecture for threat detection, sandboxing, and data protection.
- Use AES-256 adaptive encryption for email messages that contain sensitive information.
- Use RSA-2048 digital signatures to make sure that the information is real and hasn't been changed.
- Use regex rules, domain reputation checks, and safe-link validation to make phishing detection better.
- Add hybrid malware scanning (signature + behavior analysis).

- Attach a sandbox module for analyzing suspicious attachments.
- Add a self-destruct email feature for messages that need to be sent quickly.
- Do comparative testing with email services that are already out there, like Gmail and Outlook.
- Make a simple CLI or dashboard that shows alerts, logs, and risk scores.

1.5 Scope of the Project

Scope

The project focuses on building a lightweight and user-friendly email security tool designed for individuals and small organizations. It includes:

- **Email Retrieval:** Use secure IMAP (SSL/TLS) integration to get and keep an eye on new emails.
- **Threat Detection:** Using open APIs, YARA rules, heuristic checks, regex-based phishing detection, suspicious URL validation, and domain reputation lookups to look for threats on multiple levels.
- **Sandboxing:** Running suspicious attachments in a safe way with Sandbox or a small Cuckoo setup.
- **User Interface:** A attractive GUI dashboard or lightweight command line interface (CLI) that shows alerts, risk levels, and logs.
- **Safe Logging:** Local logs that are scanning and basic summary reports.

Limitations: The tool might have limitations, like the email clients it works with, the types of files it can handle, and the number of emails it can process at once. Future improvements can fix these problems.

1.6 Chapter Summary

Chapter 1 talked about the project's purpose, stakeholders, motivation, challenges, importance, goals, objectives, and scope. It shows that there is a need for a lightweight, multi-layered email security tool that is good for individuals and small businesses. This tool would fill the gap between basic protection and complex enterprise systems.

Chapter 2:

Literature/Market Survey

Chapter 2:

Literature / Market Survey

2.1 Introduction

Phishing, malware, spoofing, and other advanced attacks are becoming more common, which has made email security much better. Modern solutions include simple filters and systems that scan URLs, create sandboxes, and analyze behavior to find more complex threats. These tools try to keep users safe from a wide range of email-based threats while still being fast and easy to use. The next sections look at the latest technologies and products in this field.

2.2 Literature Review / Technologies & Products Overview

Email Security Methods and Protocols

A strong email security system usually has a number of authentication protocols to make checks of legitimacy stronger:

- SPF (Sender Policy Framework) checks that the sender's IP address is allowed to send em ail on behalf of a domain. This greatly lowers the risk of spoofing.
- DKIM (DomainKeys Identified Mail) adds a cryptographic signature to emails, which hel ps the people who get them make sure they are real.
- DMARC (Domain-based Message Authentication, Reporting, and Conformance) works with SPF/DKIM to tell receivers what to do with messages that don't pass authentication checks.

  Research has shown that using these protocols together can make email security much bet ter, by more than 95%.

Commercial Solutions:

Several commercial email security platforms exist that combine multiple protective layers:

- Email Protection from Proofpoint Offers advanced protection against phishing, malware, business email compromise, and includes AI-powered detection and URL sandboxing. Best for big businesses. (cloudnuro.ai)
- The Mimecast Secure Email Gateway has features like sandboxing, scanning URLs and attachments, continuity, and archiving. (SCM Galaxy)
- Check Point Harmony Email & Collaboration Cloud-first solution with AI-driven phishing and malware defenses, plus integration with collaboration apps. (eSecurity Planet)

- Cloud-native solutions like Avanan and IRONSCALES leverage API-based integration with Microsoft 365 or Google Workspace to provide phishing and malware detection with minimal impact on mail flow. (SCM Galaxy)

While these tools offer comprehensive protection, they often bring complexity, high pricing, and may be resource-intensive challenges for small businesses and individual users. (Expert Insights)

Open-Source and Traditional Tools

- MailScanner An open-source email filtering system that scans for spam, viruses, malware, and phishing by examining patterns in emails and obfuscated URLs. (Wikipedia)
- Sophos PureMessage Corporate anti-spam solution that detects malware and phishing while preventing data leakage. (Wikipedia)

2.3 Comparative Analysis

Feature	**Proofpoint**	**Mimecast**	**Check Point Harmony**	**Open-Source Tools**	**Proposed Tool**
Phishing Detection	AI-driven, URL scanning	URL & attachment sandboxing	AI + threat intel	Rule-based	Regex + Virus Total + Reputation
Sandbox Analysis	Yes	Yes	Limited	No	VM-based sandbox
Header Authentication (SPF/DKIM)	Basic	Basic	Integrated	Manual / Setup	Built-in
Encryption	Basic	Basic	Yes	Depends	AES-256 + RSA-2048
GUI / Dashboard	Yes	Yes	Yes	Community interface	Yes + Risk Score
Suitable for SMB	Enterprise focus	Enterprise focus	Mid to large	Suitable	Designed for lightweight use

Table 2.3 Comparison Table

Key insights from comparison:

- Enterprise tools like Proofpoint and Mimecast offer advanced AI and sandboxing, but they are usually more expensive and take a lot of time to set up.. (Expert Insights)
- Open-source tools are light and can be changed to fit your needs, but they might not be able to detect threats in real time. (Wikipedia)
- Many commercial solutions support encryption and authentication, but few put together sandboxing, phishing heuristic analysis, header checks, and encryption in a small package that works for small users.

2.4 Research Gaps

Despite the availability of various email security solutions, several gaps remain:

- **Phishing detection weaknesses:** Many solutions cannot effectively identify sophisticated phishing techniques, including fake domains or look-alike URLs.
- **High cost and complexity:** Advanced enterprise-grade tools with multi-layered security are often too expensive or technically complex for individual users and small businesses.
- **Lack of integrated multi-layered security:** Existing market solutions usually focus on a single aspect, such as spam filtering or malware detection, leaving users exposed to other risks.
- **Insufficient data protection:** Sensitive email content is often not encrypted or authenticated properly in lightweight systems, leaving data vulnerable to interception or tampering.

These gaps highlight the need for a lightweight, modular, and multi-layered email security tool that provides malware and phishing detection, sandboxing, encryption, and real-time alerts without overloading system resources.

2.5 Problem Statement

Phishing, spoofing, malware attachments, and sensitive data leaks are just a few of the modern cyber threats that can easily get into email systems. Most email providers use signature-based detection, which only finds known threats and often misses new or changed attacks. Most systems don't do a good job of detecting phishing because attackers use social engineering, fake domains, and bad links to get around regular filters. Small businesses and individual users are especially at risk because they can't afford or get to advanced security tools. Also, lightweight

systems don't often use strong encryption or digital signatures to keep sensitive information like passwords, financial records, and private business information safe.

The core problem is the absence of a lightweight, multi-layered email security solution that can:

- Detect advanced threats including zero-day malware and phishing attacks,
- Ensure confidentiality and authenticity of sensitive information through AES encryption and RSA digital signatures,
- Remain practical and resource-efficient for individuals and small organizations.

2.6 Chapter Summary

This chapter looked at current email security technologies and talked about their pros and cons. Gmail and Outlook are two well-known email services that offer basic protection against spam and malware by scanning for signatures. However, they often don't work against more advanced or unknown threats. Literature and product surveys show that there is a need for lightweight, integrated solutions that protect data, stop phishing, detect malware, and sandbox. The identified research gaps, such as inadequate threat coverage, ineffective phishing detection, excessive costs, and insufficient multi-layered security, necessitate the creation of a Lightweight Email Security Tool utilizing Sandbox technology. The next chapters will talk about the system requirements, design, proposed solution, and plan for putting it into action to solve these problems.

Chapter 3:

Requirements and System Design

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Requirements and System Design

3.1 Introduction

This chapter explains how to use Sandbox to design and build the Lightweight Email Security Tool. It talks about system architecture, both functional and non-functional requirements, hardware and software needs, and threat modelling. The chapter gives you a plan for setting up a strong, light, and safe email scanning system.

3.2 System Architecture

The system has a modular, multi-layered design:

1. **Email Input Module:** Records emails that come in and go out.
2. **Phishing Detection Module:** Uses regex patterns, domain reputation checks and Virus Total scan to look at the content and links in emails.
3. **Header Verification Module:** Checks the authenticity of email headers (Sender IP, SPF, DKIM).
4. **Attachment Analysis Module:** Sends files that look suspicious (PDF, DOCX, EXE, ZIP) to a virtual machine-based sandbox so that malware can be found.
5. **Encryption Module:** Uses AES-256 to encrypt email content and RSA-2048 to sign emails to make sure they are safe.
6. **The GUI/Dashboard Module:** shows alerts, logs, and risk scores for each email.

Workflow Overview:

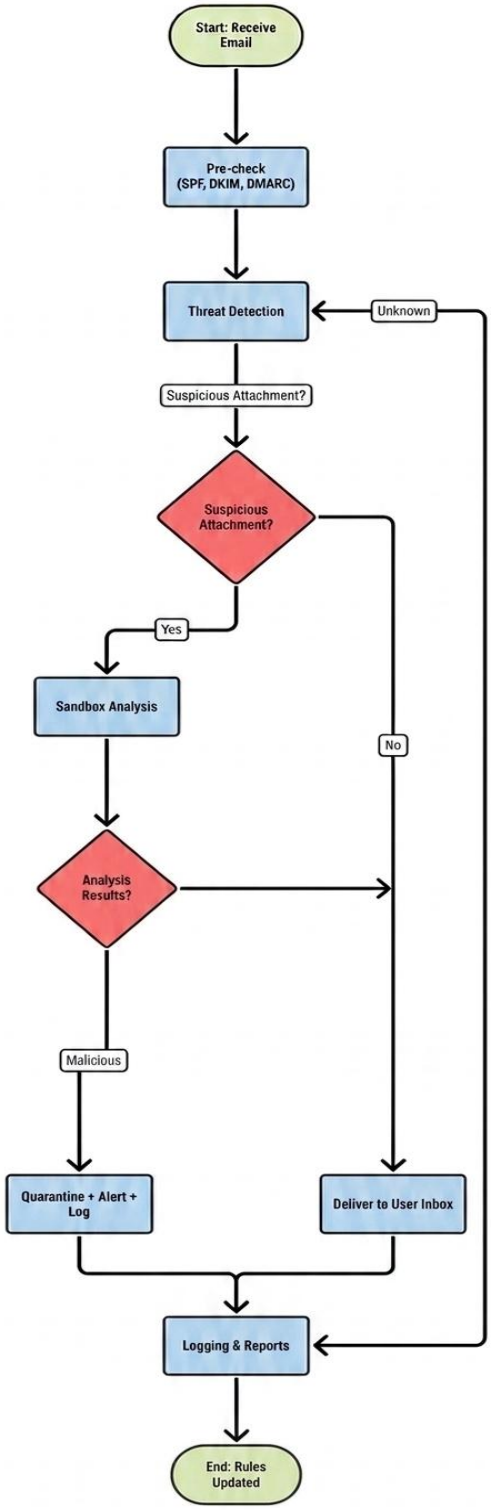


Figure 3.2 Workflow

3.3 Functional Requirements

The system must:

- Check all emails that come in and go out for phishing links.
- Check the email headers (SPF, DKIM, Sender IP) to make sure the sender is real.
- Look at attachments (PDF, DOCX, EXE, ZIP) in a VM-based sandbox.
- Encrypt the content of emails and add digital signatures.
- Give users a GUI or dashboard with alerts and risk scores that are updated in real time.
- Keep records of all scanned emails for auditing reasons.

Req ID	Requirement	Description	Priority
FR1	Email Scanning	Scan incoming/outgoing emails for threats	High
FR2	Phishing Detection	Detect malicious links using regex & reputation	High
FR3	Header Verification	Validate SPF, DKIM, sender IP	High
FR4	Attachment Analysis	Execute files in sandbox environment	High
FR5	Encryption	Encrypt email using AES-256	Medium
FR6	Digital Signature	Ensure integrity using RSA-2048	Medium
FR7	Alert System	Notify users with risk scores	High
FR8	Logging	Store scan results and logs	Medium

Table 3.3 Functional Requirements

3.4 Non-Functional Requirements

Performance: Scanning shouldn't slow down the delivery of emails too much.

Usability: The GUI must be easy to use and show clear alerts.

Reliability: The system must be able to handle more than one email at a time without crashing.

Security: AES-256 encryption and RSA-2048 digital signature are both strong.

Portability: works with most email clients (at first for SMTP/IMAP).

Category	Requirement	Description
Performance	Fast Processing	Minimal delay in email delivery
Usability	User-Friendly UI	Easy dashboard/CLI
Reliability	Stability	Handle multiple emails simultaneously
Security	Strong Encryption	AES-256 and RSA-2048
Scalability	Moderate Load	Suitable for small organizations

Table 3.4 Non Functional Requirements

3.5 Design Diagrams

3.5.1 Architecture Diagram:

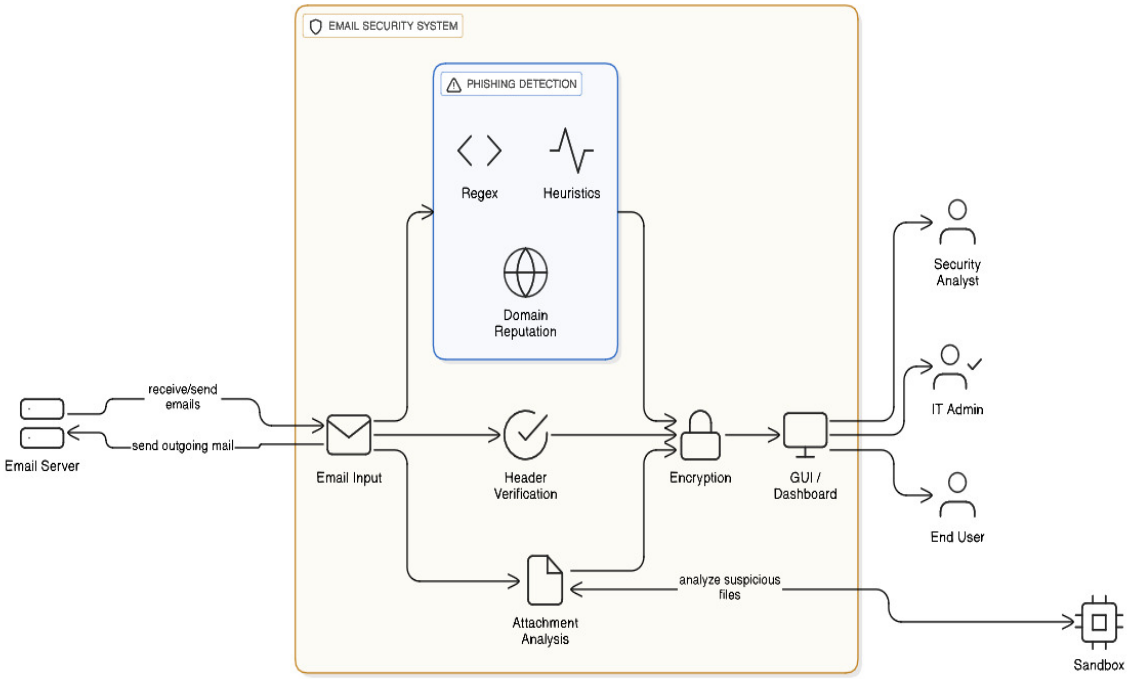


Figure 3.5.1

3.5.2 Data Flow Diagram (DFD):

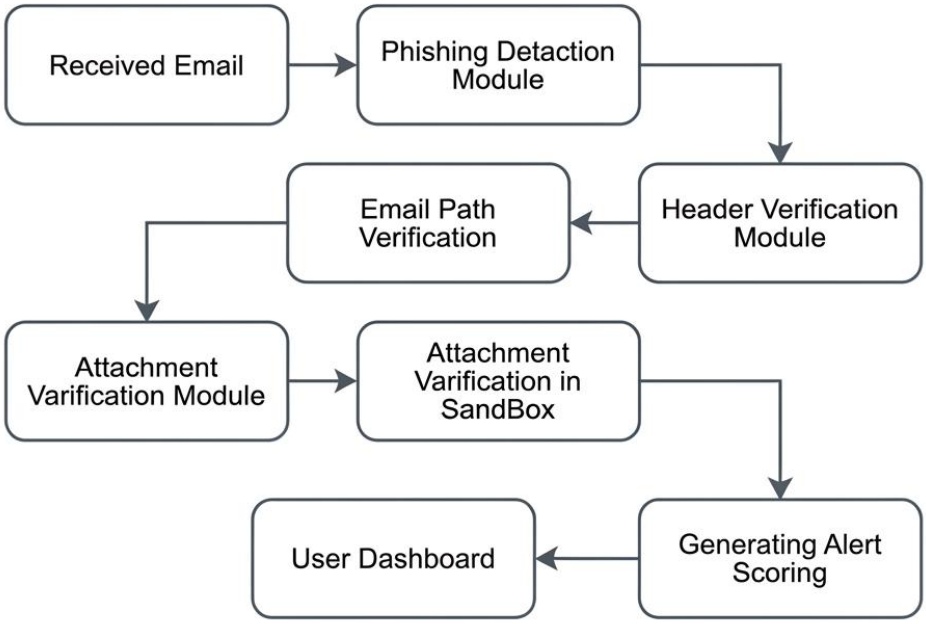


Figure 3.5.2

3.5.3 Sequence Diagram:

User sends or receives email, → system scans it → Found phishing? → Attachment sent to the sandbox → Results to GUI → Email sent and encrypted

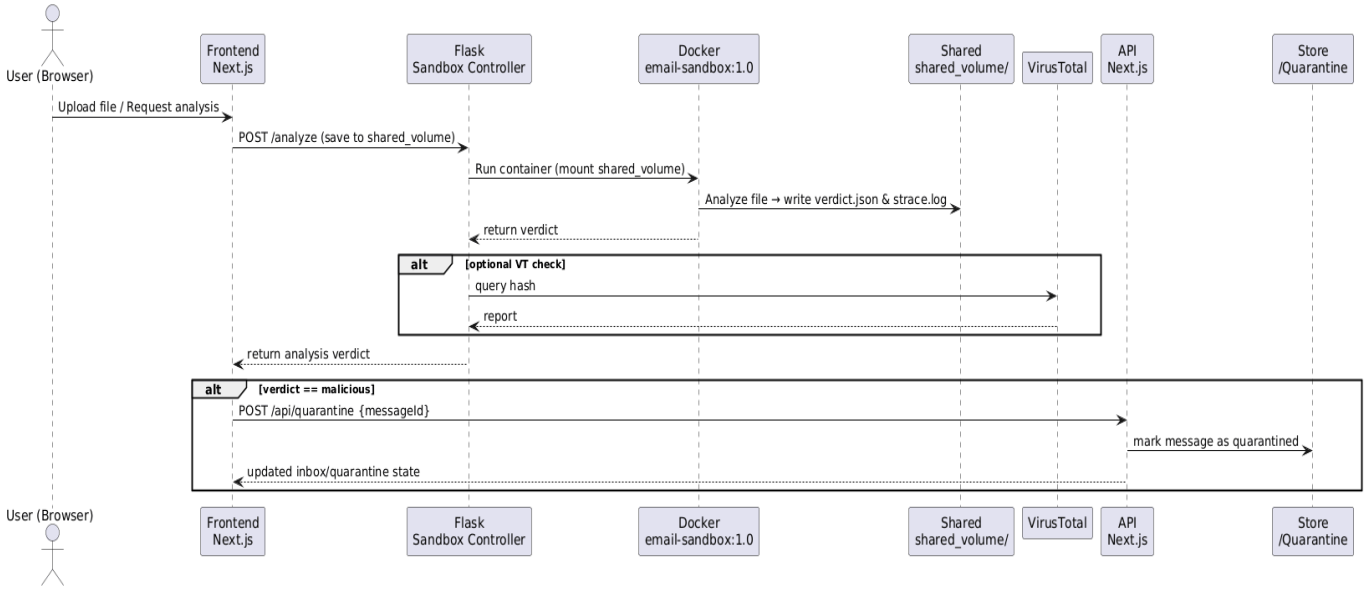


Figure 3.5.3

3.6 Hardware and Software Requirements

Hardware:

- **Processor:** Minimum of a dual-core CPU (Intel i3 or similar)
- **RAM**: 2 to 4 GB (enough for analysing attachments in a sandbox without slowing down the system)
- **Storage:** 20 to 50 GB of free space (enough for emails, logs, and temporary sandbox files)
- **Network:** Standard internet connection for IMAP/SMTP

Software:

- Operating System: Windows 10/11 or Linux
- Programming Language: Python
- Libraries: flask, flask-login, flask-wtf, python-dotenv, cryptography, email-validator, Werkzeug, virus total-python,
- Database: filesystem JSON store
- Virtual Machine: VMware for sandbox execution

Requirement Type	Component	Specification
Hardware	Processor	Minimum Dual-Core (Intel i3 or equivalent)
Hardware	RAM	2–4 GB
Hardware	Storage	20–50 GB free space
Hardware	Network	Stable internet for IMAP/SMTP
Software	OS	Windows 10/11 or Linux
Software	Language	Python
Software	Libraries	Flask, Cryptography, email-validator, Werkzeug
Software	Database	JSON-based file system
Software	Sandbox	VMware / Virtual Machine

Table 3.6 System Requirements

3.7 Threat Scenarios

- **Phishing attacks:** Emails with bad links that try to steal your login information.
- **Malware attachments:** Files that try to break into the host system.
- **Spoofing:** Sending fake sender addresses that don't go through authentication.
- **Data leakage:** getting into email content without permission.

Threat Type	Description	Detection Method	Mitigation
Phishing	Fake links & domains	Regex + URL validation	Block / Alert
Malware	Malicious attachments	Sandbox execution	Quarantine
Spoofing	Fake sender identity	SPF/DKIM checks	Reject email
Data Leakage	Unauthorized access	Encryption	Secure storage

Table 3.7 Threat Mapping Table

3.8 Threat Modeling Techniques

- **STRIDE Model** lists threats that fall under the following categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.
- **Attack Trees:** Look at possible ways for malware or phishing attacks to get in.
- **Sandbox Isolation:** Keeps bad code from running on the main system.

3.9 Threat Resistance Model

- **Multi-layered detection** (phishing, header, sandbox) stops failures from happening in one place.
- **AES-256 encryption** keeps things private, and RSA-2048 signatures make sure they are correct.
- **Continuous logging** makes it possible to trace things back and look at what happened after an incident.

3.10 Chapter Summary

This chapter outlined the specifications and architecture of the proposed email security tool. It talked about the system architecture, the functional and non-functional requirements, the threat scenarios, and the ways to deal with them. The groundwork has been laid for the next chapter's implementation and testing.

Chapter 4:

Proposed Solution

Chapter 4

SYSTEM DESIGN

4.1 Introduction

The Lightweight Email Security Tool with Sandbox that is being suggested will protect individual users and small businesses in many ways. It includes malware and phishing detection, sandboxing of attachments that look suspicious, AES-based encryption, RSA digital signatures, and alerts for users in real time. The system is made to be lightweight, modular, and use as few resources as possible, so that email communication is safe without slowing down performance.

4.2 Proposed Model

The system is a hybrid experimental model with a modular, multi-layered architecture that makes sure all emails are safe. There are four main parts to it:

- **Threat Detection Module:** This module uses regex-based phishing detection, domain reputation checks, and safe-link validation along with signature-based and Virus Total malware scanning. To check the sender's identity, emails are checked for their URLs and headers (SPF, DKIM, and Received-Path).
- **Sandboxing Module:** Attachments that look suspicious are run in a VM-based isolated environment to find hidden malware and changing threats.
- **Data Protection Module:** AES-256 encryption and RSA-2048 digital signatures are used to protect clean emails. The module also lets you send self-destructing emails for messages that need to be sent right away.
- **User Alerting Module:** A lightweight Dashboard shows scan results, risk scores, and alerts, giving users real-time feedback and making things clear.
  The model runs each email through a multi-stage pipeline that uses both static and dynamic analysis to find threats at different points. This layered approach offers better protection than single-layer security tools, and it is also lightweight and doesn't use up too many resources.

4.3 Data Collection

To test how well the system works, this project will collect sample emails, metadata, and attachments from both controlled and publicly available sources. Test emails include fake phishing emails, harmless corporate emails, and spam-like emails that were made automatically. We also get more datasets from security repositories that have phishing URLs, suspicious headers, and malware samples. When attachments are run in a sandbox, network logs are kept to look for things like file changes, registry changes, or outgoing connections. To keep things running smoothly, all the data that is collected is either processed in memory or temporarily stored in lightweight storage. No real user personal data is used, which is in line with ethical and privacy standards.

4.4 Data Pre-Processing

Preprocessing is done to improve and refine the dataset before it is fed into the system.
This includes:

- **Filtering and classification:** Emails are sorted into two groups like: Suspicious and Safe. This is done by looking at the headers, patterns in the content, and the attachments.
- **Making suspicious things normal:** Obfuscated links and encoded payloads are made the same so that they are scanned using Virus Total.
- **Outlier handling:** Extremely large attachments or malformed traffic patterns are retained but flagged for special handling during sandbox analysis.
- **Prioritization:** Emails that pose a high risk are handled first so that they can be found and responded to more quickly.
- **Preparing the attachment:** Files that look suspicious are taken out and set up to run in a sandbox.
- **Quarantine the Email:** The mails that are found suspicious are quarantined manually by user.

These steps before processing make the system more accurate, reliable, and efficient when it is being tested and analyzed.

4.5 Tools and Techniques

The project uses a mix of security tools, cryptographic algorithms, and analysis methods to put the proposed model into action:

- **Handling Email:** Python libraries like imaplib and email are used to get, read, and process emails from user accounts.
- **Malware Detection:** To find known and zero-day threats, a hybrid method that uses both signature-based scanning and heuristic rules is used.

- **Phishing Detection:** To find fake URLs and spoofed domains, regex filtering, domain reputation checks, and safe-link validation are used.
- **Sandboxing:** Suspicious attachments like PDFs, DOCXs, ZIPs, and executable files are run in a virtual machine (VM) that is separate from the rest of the system. This lets you watch system calls, network activity, and signs of malware.
- **Cryptography:** AES-256 symmetric encryption keeps email content safe, and RSA-2048 digital signatures make sure it is real and correct.
- **Header Verification:** Protocol-based checks like SPF, DKIM, and DMARC make sure that the sender is who they say they are.
- **Dashboard:** An user interface shows end users real-time alerts, quarantined, logs, and risk scores.

Category	Tool / Technique	Purpose
Email Handling	imaplib, email (Python)	Fetch and parse emails
Phishing Detection	Regex, Domain Reputation	Detect fake links/domains
Malware Detection	VirusTotal API, Signature + Heuristic	Identify malicious files
Sandboxing	Virtual Machine (VMware/Docker)	Safe execution of attachments
Encryption	AES-256	Data confidentiality
Digital Signature	RSA-2048	Integrity & authentication
Header Verification	SPF, DKIM, DMARC	Sender validation
Dashboard	Flask GUI / CLI	Alerts, logs, risk scores

Table 4.5 Tools & Tecnnique

4.6 Evaluation Metrics

We use metrics that look at both security effectiveness and operational performance to test the proposed email security system.

- **Accuracy:** This checks how well the system can tell the difference between phishing and real emails.
- **Detection Rate and False Positive Rate:** These measure how well threats are found while keeping the number of safe emails that are wrongly classified to a minimum.
- **Processing Speed:** This checks how quickly each module can finish scanning without slowing down email delivery.
- **Resource Usage:** Keeps an eye on how much RAM, CPU, and storage the system uses to make sure it stays light and fast.

- **Security Robustness:** This checks how well the sandbox finds bad behaviour and how well AES encryption and RSA signatures keep sensitive information safe.
- **Usability:** Looks at how clear the GUI/CLI dashboard is, how easy it is to understand alerts, and how severity scores are shown.

All of these metrics show that the system is reliable, useful, and good for small-scale users, making sure that the tool works well and doesn't waste resources.

4.7 Chapter Summary

This chapter gave a full picture of the suggested solution, including its modular architecture, operational workflow, and the techniques that make it work. The system combines phishing detection, header validation, sandbox-based malware analysis, and encryption into a single process that makes email safer. Data collection and preprocessing make sure that the inputs are structured and useful for accurate testing. The use of cryptographic methods and analysis tools together makes the system as a whole stronger. The tool meets modern cybersecurity standards based on metrics like accuracy, detection rate, false positives, processing speed, resource usage, and usability. The next chapter will talk about how this lightweight email security tool was put into use, tested, and validated to see how well it works, how efficient it is, and how easy it is to use in real life.

Chapter 5:

Implementation and Testing

Chapter 5: Implementation and Testing

5.1 Security Properties Testing

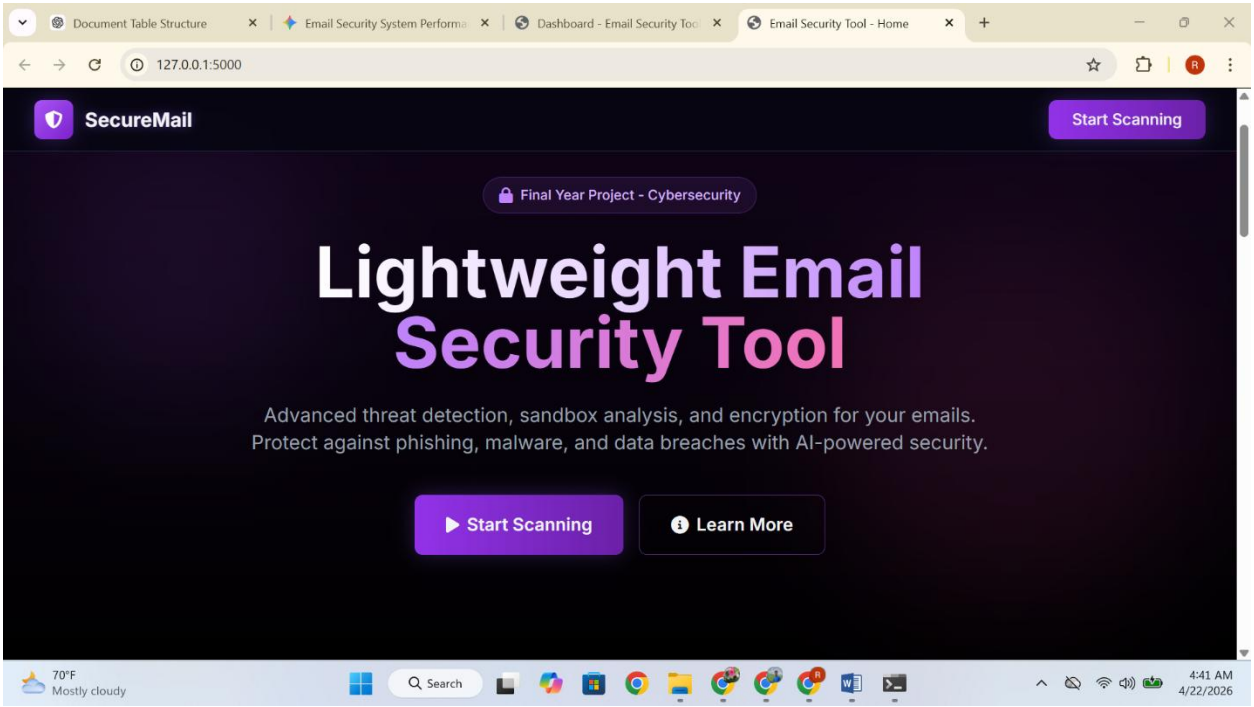
Security testing checks that the email security system is reliable, accurate, and strong. We test the malware detection module against real and fake threats to make sure it can correctly find malicious content. We also test the phishing detection module using fake URLs, look-alike domains, and malicious email bodies. We keep an eye on sandbox execution to make sure that attachments are kept separate and that any bad behavior is caught without affecting the host system. AES-256 encryption checks that privacy is protected, so that sensitive emails and attachments stay safe. Integrity testing makes sure that messages don't change while being scanned, encrypted, or sent. To stop spoofing, SPF, DKIM, and DMARC header checks are used to check authentication. To stop senders from denying the origin of a message, RSA-2048 digital signatures are used to enforce non-repudiation. This thorough testing shows that the system does a good job of protecting against both basic and more advanced email threats.

5.2 System Setup

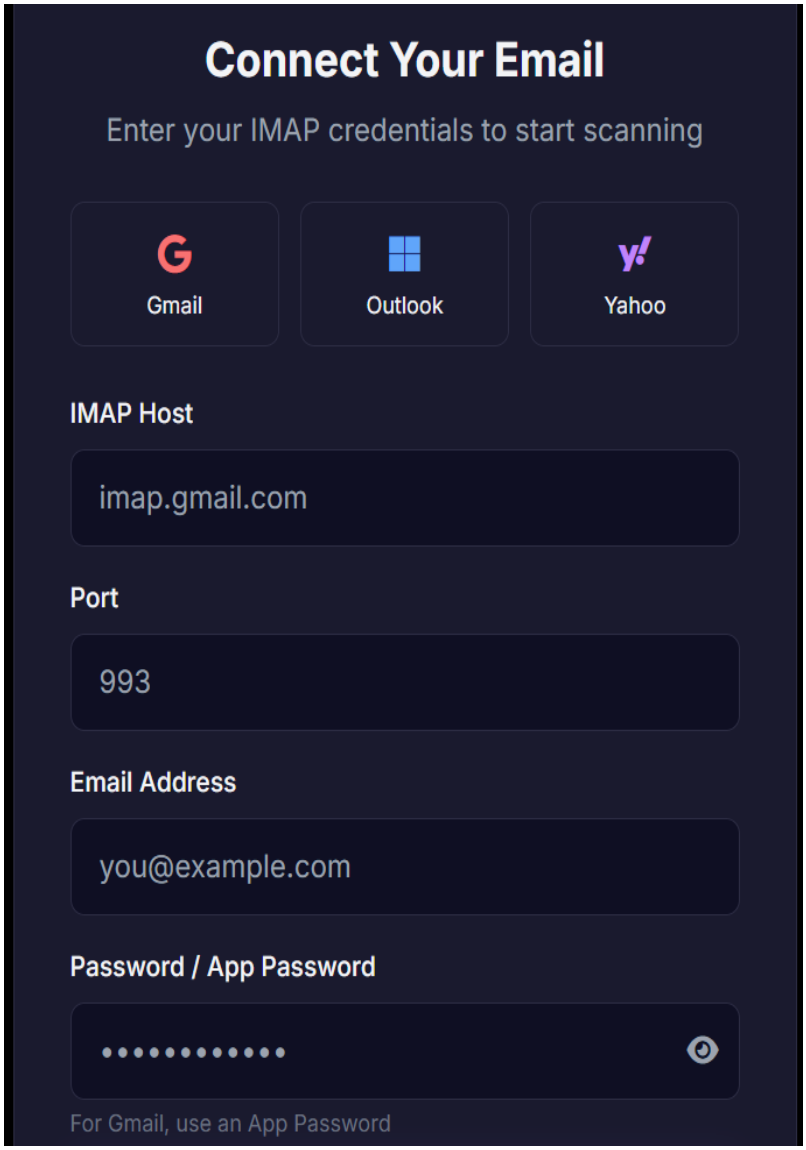
To make sure that all modules work safely, the system is set up in a controlled, light, and resource-efficient way. Some of the most important parts are:

- **Programming Environment:** The core modules and the CLI/dashboard interface are written in Python 3.x.
- **Email Handling:** Libraries like imaplib and email take care of getting and parsing emails.
- **Sandbox Environment:** Attachments are run in separate virtual machines or containerized environments (like Docker) for malware behavioral analysis. There are strict network restrictions to stop the malware from spreading by accident.
- **Encryption and Signatures:** AES-256 keeps email content private, and RSA-2048 is used for safe key exchange and digital signatures.
- **Integration and APIs:** Well-defined APIs let modules talk to each other without any problems. This lets emails be sent to the sandbox, scan results be retrieved, and alerts be sent in real time.
- **Logging and Monitoring:** Metadata, behavioral patterns, and classification decisions are saved to help with debugging, performance evaluation, and keeping an eye on compliance.

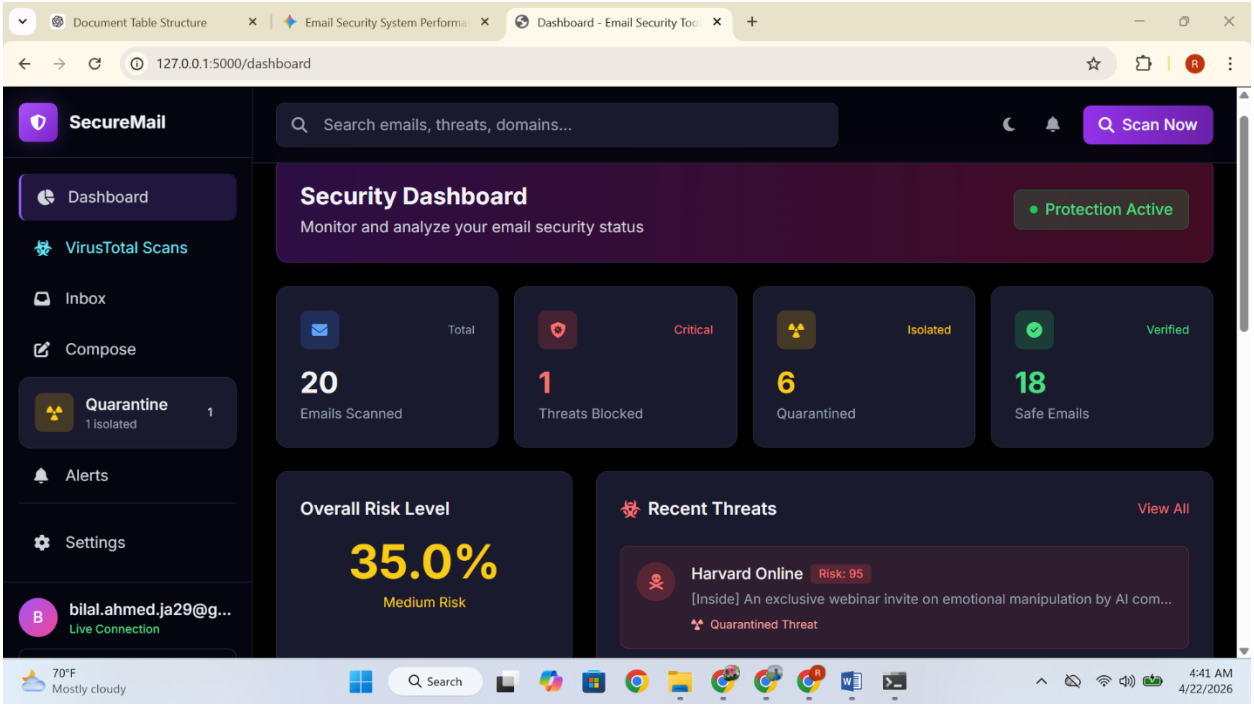
This organized setup makes sure that the system works reliably, safely, and quickly during testing and evaluation.



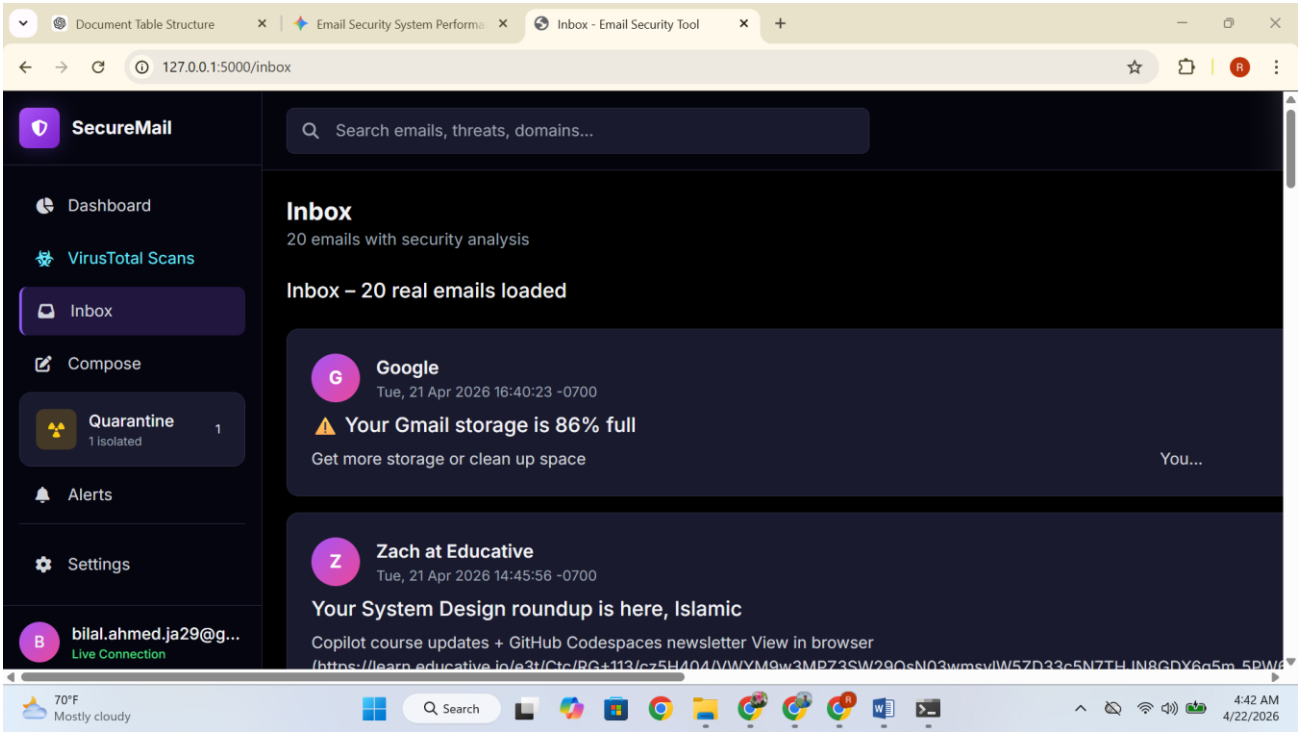
Screenshot: Initial landing page of the email security system interface



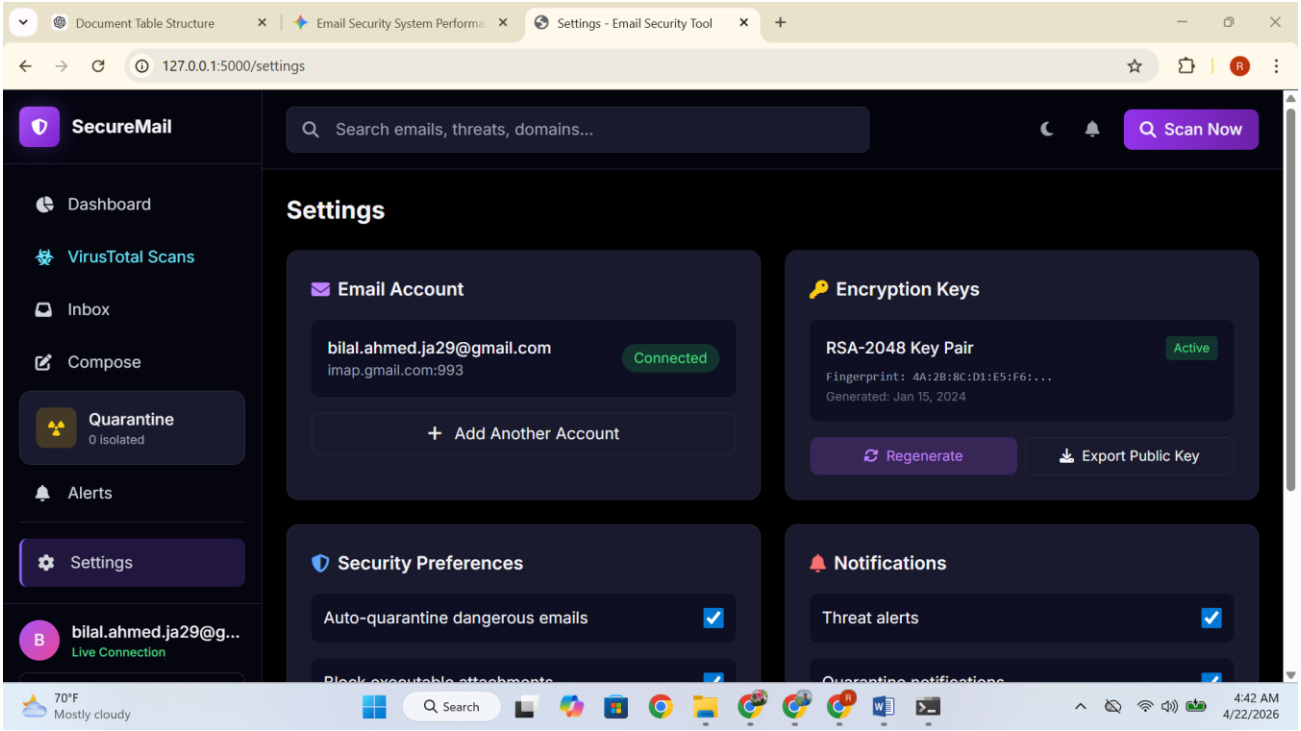
Screenshot: User authentication interface for secure system access.



Screenshot: Central dashboard displaying system status, alerts, and risk overview



Screenshot: Interface showing received emails for analysis and monitoring

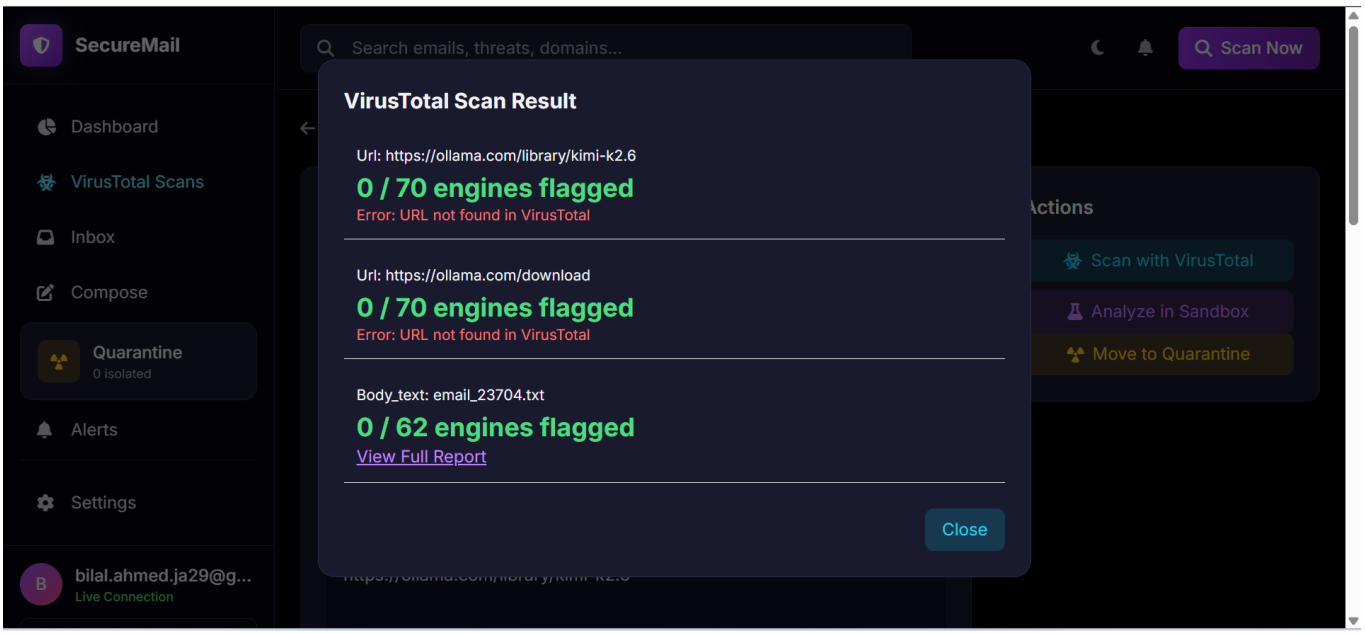


Screenshot: Configuration panel for system preferences and security options.

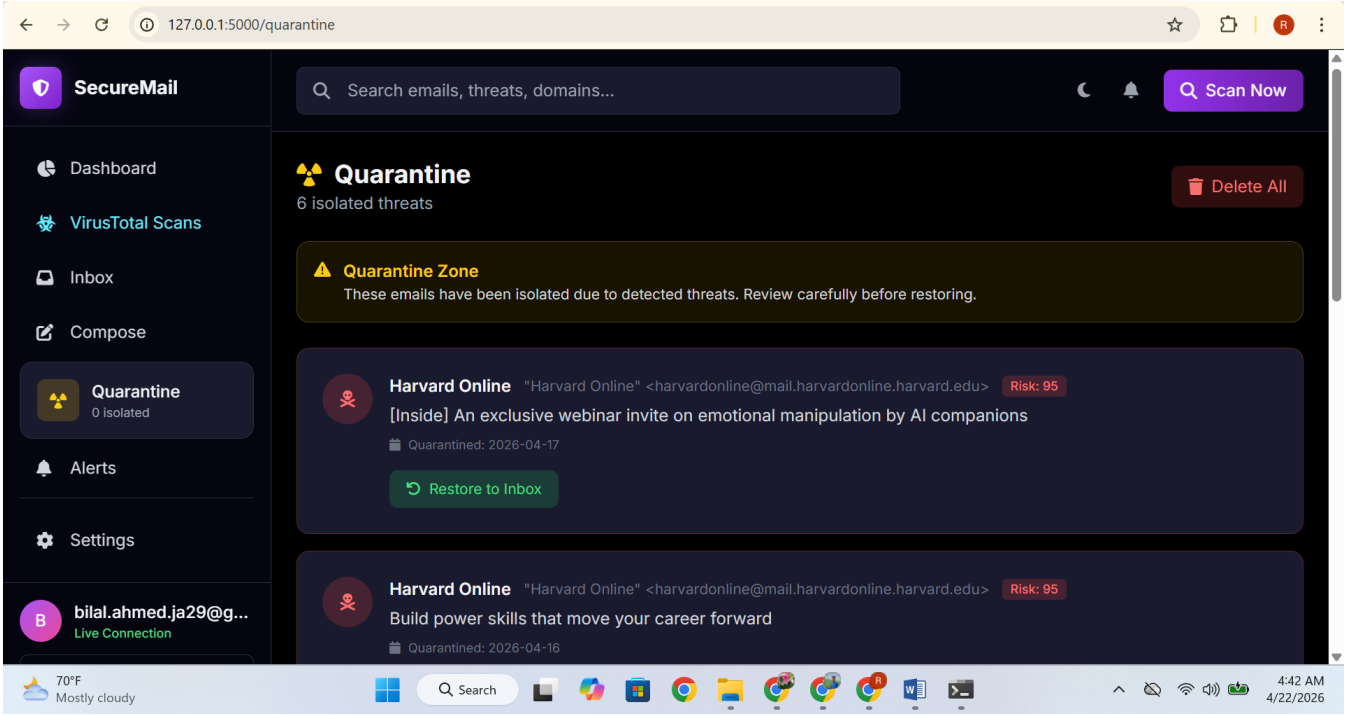
5.3 System Integration

System integration makes sure that all the modules work together as one email security solution. The main modules for Threat Detection, Sandboxing, Data Protection, and User Alerting talk to each other through secure inter-process messaging and predefined APIs. Before being sent, emails that are retrieved through IMAP are first checked for phishing and malware. Attachments that look suspicious are sent to the sandbox to run separately.

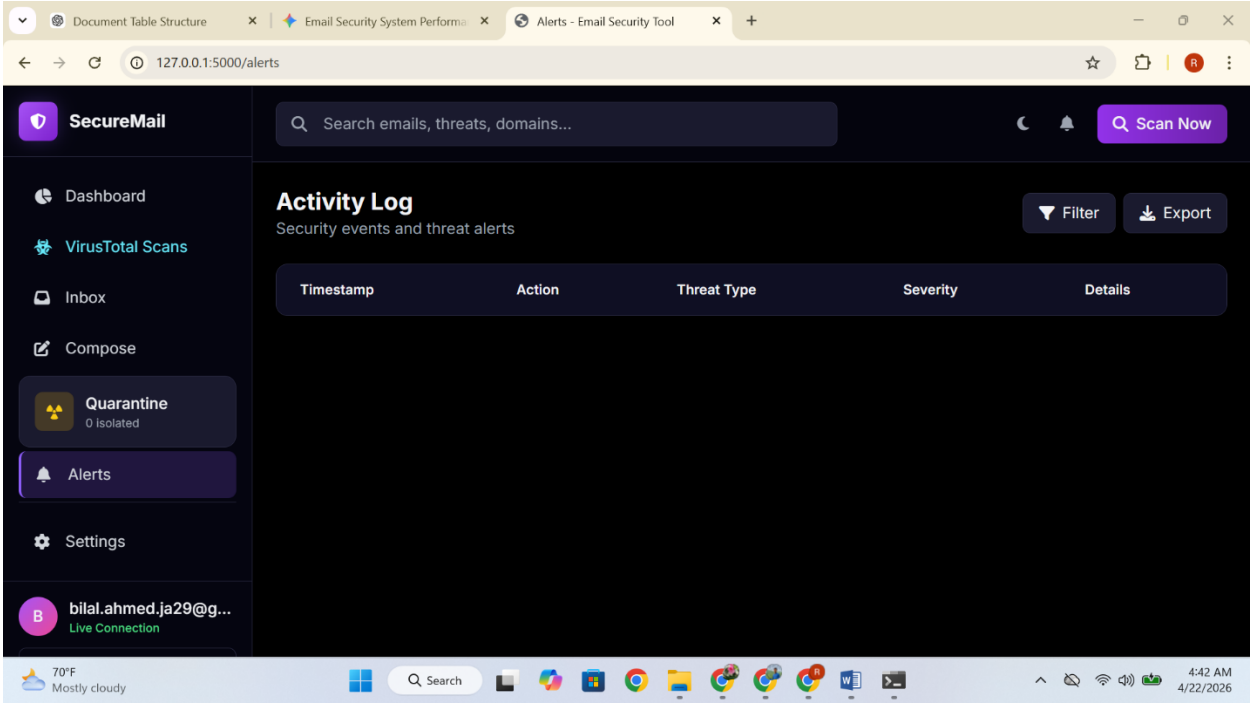
A central database stores all scan results, risk scores, alerts, and logs. This lets the GUI/CLI dashboard show information in real time. Communication protocols and shared data structures make sure that each module gets the right inputs and gives the right outputs. This keeps the pipeline consistent, safe, and in sync. This integrated setup makes sure that the system works as a smooth, automated workflow that can find and stop threats at different points without any problems or extra work.



Screenshot: Email scanning process analyzing content, links, and attachments



Screenshot: Isolated storage of suspicious or malicious emails detected by the system.



Screenshot: Alert module indicating no threats detected during the current session.

5.4 Reverse Engineering and Malware Analysis (Optional)

The sandbox uses dynamic monitoring of system calls, file operations, and network activity to look at suspicious attachments. This lets you find hidden malware behaviours like payloads that are hard to read or code that changes shape. Reverse engineering techniques help us figure out how malware works and make heuristic detection rules better.

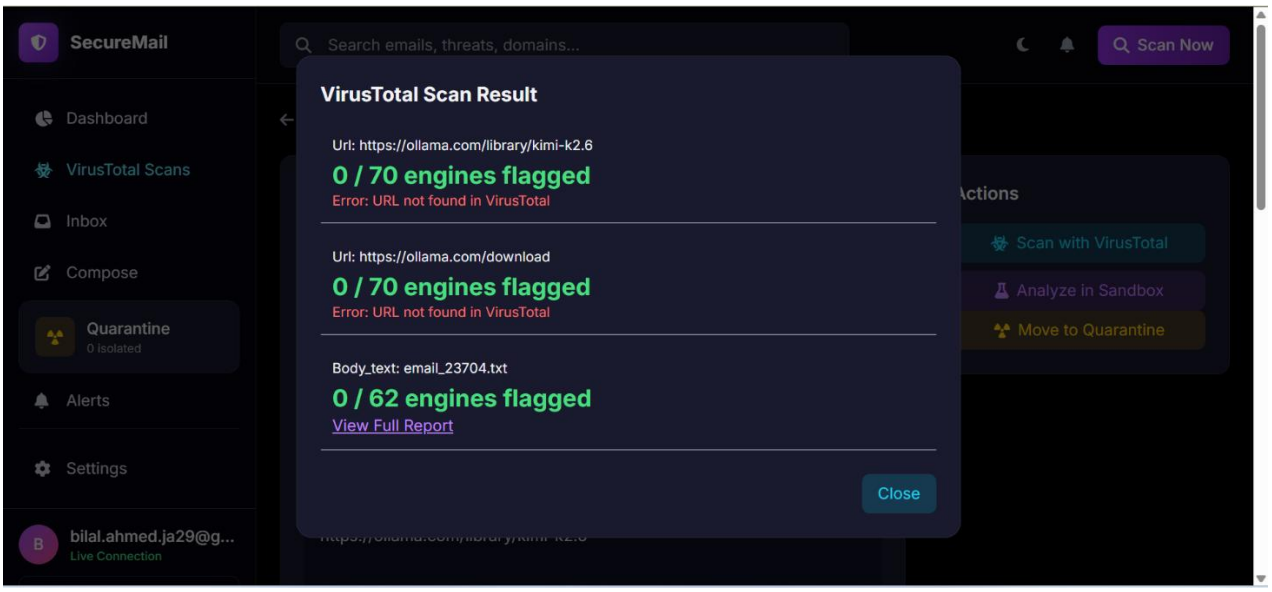
5.5 Test Cases

Test cases are designed to evaluate the performance, accuracy, and functionality of each module in a controlled and repeatable manner. Scenarios include:

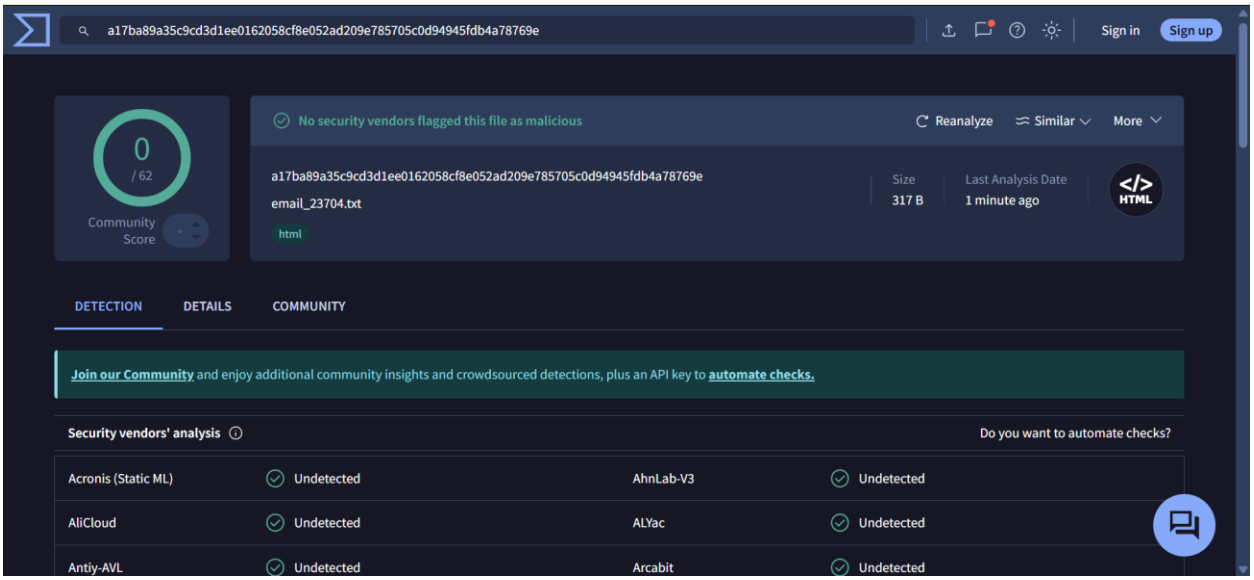
- Receiving clean emails to verify correct handling.
- Detecting phishing URLs and look-alike domains.
- Identifying spoofed sender addresses and malformed headers.
- Executing infected attachments in the sandbox (.exe, .docx, .pdf, .zip).
- Applying encryption to legitimate messages.

Functional tests check that each module does what it is supposed to do without making any mistakes. Performance tests check the speed of processing, response time, and overall throughput.

Negative test cases make sure that safe messages are not incorrectly classified, which lowers the number of false positives. Stress testing checks how well the system can handle a lot of emails, multiple attachments, and scanning operations that happen at the same time. The results of these tests give a full picture of how reliable, accurate, and strong the system is.



Screenshot: Summary report showing overall scan results and risk evaluation.



Screenshot: Detailed Virustotal report providing in-depth malware analysis of scanned files.

5.6 Results and Discussion

Tests show that the suggested email security system works well in all of its parts. The phishing detection engine was very accurate, finding malicious URLs, look-alike domains, and misleading email content. The malware detection and sandbox module caught all malicious actions while they were happening, such as changing files, changing the registry, and suspicious network activity, without affecting the host system. This shows that it was able to effectively isolate the malware.

SPF, DKIM, and DMARC header validation always found fake senders and mail servers that weren't allowed to send mail. Tests on encryption showed that AES-256 and RSA-2048 were used correctly, which kept the communication process private and safe.

Performance measurements further confirm the tool’s lightweight design: average RAM consumption remained below 4GB, storage usage stayed under 20GB, and processing delays were minimal, making the system viable for individual users and small organizations. False positives stayed low, so real emails weren't blocked for no reason.

Overall, the results show that the system's multi-layered architecture provides strong security, good performance, and easy use in real-world settings.

Test Case	Description	Expected Result	Actual Result	Status
TC1	Phishing Email	Detected as phishing	Detected correctly	✅ Pass
TC2	Malware Attachment	Sandbox detection	Malware detected	✅ Pass
TC3	Spoofed Sender	Header failure detection	Detected spoofing	✅ Pass
TC4	Benign Email	No flagging	Delivered normally	✅ Pass
TC5	Spam-like Email	Mark suspicious	Flagged correctly	✅ Pass

Table 5.6 Testing Results

5.7 Best Practices and Coding Standards

To make sure that all modules are easy to maintain, reliable, and secure, the email security system was built using strict coding and engineering standards.

5.7.1 Code Validation:

A mix of automated and manual review methods are used to check the code:

- **Static code analysis** tools like pylint and flake8 look for syntax errors, variables that aren't being used, functions that aren't safe, and code that doesn't follow Python best practices.
- **Logical validation** makes sure that control flows, condition checks, and exception handling work as they should in all modules, but especially in parts that are important for security.
- **Unit tests** are written for the core functions of email parsing, URL scanning, header analysis, sandbox routing, and encryption to make sure they always work the same way.
- **Peer review and manual inspection** are used to find structural problems and make sure that security-sensitive code paths are correct.

5.7.2 Development Practices:

To keep the codebase clean and easy to grow, we follow industry-standard development practices:

- **Modular architecture** makes sure that each part of phishing detection, sandboxing, encryption, and header validation can be tested, reused, and kept up to date on its own.
- **Comprehensive documentation** is given to all classes, functions, and API endpoints to make sure they can be maintained for a long time.
- **Git** keeps version control in check by using clear commit messages, separate feature branches, and organized merge processes.
- **Error handling and logging** are always done the same way so that there are no silent failures and it is easy to find problems when debugging and testing.

5.7.3 Secure Coding:

The whole system follows security-focused engineering principles:

- **Strict input validation** stops malformed headers, malicious URLs, and unsafe filenames from causing vulnerabilities.
- **Safe file handling** means that attachments are only opened in sandboxed or temporary places, which keeps them from being accidentally run or seen.
- **Using AES-256 and RSA-2048** with the right padding, storing keys securely, and not using old libraries keeps cryptographic correctness.
- **Explicit sanitization and controlled** workflows make sure that common attacks like injection, unsafe deserialization, command execution, and insecure API calls don't work.

5.8 Chapter Summary

This chapter went into great detail about how to fully implement, integrate, and test the lightweight email security system. Phishing detection, header analysis, sandbox-based attachment inspection, encryption, and user alerting were all put into place, checked, and tested in a controlled setting. Testing for security properties showed that the system keeps confidentiality, integrity, authentication, and isolation during all operations.

A wide range of test cases, such as phishing emails, fake headers, and malware attachments, showed that the system had a high detection rate, few false positives, and overall light performance. The sandbox environment was able to catch bad behaviour without affecting the host system, and the AES and RSA modules made sure that data was handled safely. The chapter also stressed the importance of following secure coding standards and best practices for development to make sure that the code is stable and can be maintained for a long time.

The next chapter will give the project's final conclusion, as well as its most important results and suggestions for how to make it better in the future.

Chapter 6:
Conclusion and Future Work

Chapter 6: Conclusion and Future Work

6.1 Introduction

This chapter looks at the project's overall results and how well the lightweight email security tool works in real life. It shows how well the system works, points out its flaws, and talks about how useful it is for small-scale users. The section also talks about possible changes that could make the system more accurate, faster, and better able to deal with new email-based threats.

6.2 Achievements and Improvements

By combining multi-layered detection and sandbox-based analysis, the Email Security Tool fixes major flaws in traditional email protection systems. The most important things that were done are:

- **Detecting Phishing and Malware:** The system can reliably find phishing emails and harmful attachments by using a combination of static analysis, URL inspection, and controlled sandbox execution. Files that look suspicious are separated and run in a safe way. This makes it possible to find zero-day malware and advanced evasion techniques.
- **Real-Time Threat Scanning and Alerts:** All incoming emails are checked right away for strange patterns, harmful links, and harmful payloads. The system sends out alerts and risk scores right away, which cuts down on response time and makes threats easier to see for the user.
- **Header and IP Reputation Analysis:** We check the sender's authenticity by looking at the header, SPF/DKIM checks, and IP reputation scoring. This stops fake senders, mail servers that aren't authorized, and high-risk origins from getting around the system.
- **Verified Performance Through Testing:** Full tests with fake phishing campaigns, malware samples, and strange email structures show that the detection is always correct and consistent.

Sandbox logs give you a lot of information about how the system behaves, which shows that it is reliable even when it is under attack.

6.3 Critical Review

Even though the Email Security Tool has some good points, it still has some big problems that affect how well it works and how long it will last:

- **Not being able to look at encrypted attachments:** The system can't look at files that are encrypted or password-protected, which means that malware hidden in encrypted attachments can completely avoid being analyzed. This is a big blind spot that makes it harder to find threats overall as encrypted communication becomes more common.
- **Not very good at dealing with new threats:** Most detection logic is based on rules that have already been set, static signatures, and how things behave in a sandbox. The tool might not be able to find complex, fast-changing phishing campaigns or advanced malware that changes its behavior to avoid sandbox triggers if it doesn't use machine learning or get real-time updates on threats.
- **Possible Problems with Scalability:** The current design works well for one person or a small group of people, but it might have trouble with a lot of emails. Repeated sandbox executions and high-frequency attachment scanning use more resources, which slows things down. The system would need distributed sandboxing, load balancing, or better resource scheduling in order to scale well.

6.4 Future Recommendations

To make the Email Security Tool more effective, flexible, and scalable, the following changes are suggested:

- **Analysis of Encrypted Attachments**
  Add secure decryption methods or work with authorised user credentials to let people safely look at encrypted and password-protected attachments. This will fill in a big gap in your knowledge and make it easier to find threats in general.
- **Detecting threats that change over time**
  Use machine learning models or real-time threat intelligence feeds to find phishing campaigns and polymorphic malware that are changing. Dynamic learning will make it easier to find advanced and previously unknown attacks.
- **Architecture of a scalable sandbox**
  Use distributed or cloud-based sandbox environments to handle a lot of email traffic quickly and easily. Load balancing and better resource allocation will keep performance steady even when a lot of people are using it.
- **Better reporting and analytics**
  Make dashboards and automated reporting tools that give you more information about attack trends, sandbox outcomes, and email security metrics. This will help administrators make quick, smart choices.
- **Regular updates and maintenance**
  Set up automatic update systems for threat signatures, sandbox settings, and encryption protocols so that the system can keep working against new threats without needing a lot of manual work.

6.5 Chapter Summary

This chapter gave a brief overview of the Email Security Tool's overall results when used with Sandbox. The system was very good at finding phishing emails, harmful attachments, and suspicious IP addresses in real time. The sandbox execution made it possible to safely and privately look at possible threats.

Some of the main problems are that encrypted attachments can't be checked, the system can't quickly adapt to new threats, and it might not work well when there are a lot of emails. To make the system more resilient, scalable, and effective, it is suggested that future improvements include secure analysis of encrypted emails, the addition of adaptive threat detection, the use of distributed sandboxing, and the ability to work with other security solutions. By making these changes, the tool will continue to be a strong and reliable defence against new email-based cyber threats, which is important in a world where cybersecurity is always changing.

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Appendix A:

Appendices

Appendix A: Sample Emails for Testing

Test Case ID	Email Type	Description	Expected Outcome
TC1	Phishing	Email containing a malicious link mimicking a banking site	Flagged as phishing
TC2	Malware	Email with `.exe` attachment	Sandbox executes and detects malware
TC3	Spoofed Sender	Email claiming to be from trusted domain but failing SPF/DKIM	Flagged as spoofed
TC4	Benign	Normal email with harmless attachments	Delivered without warning
TC5	Spam-like	Automatically generated email with spam-like text	Flagged as suspicious

Appendix B: Sandbox Setup and Configuration

Virtual Machine Configuration

- RAM: 4 GB
- CPU: 2 Cores
- Storage: 20 GB
- OS: Lightweight Linux distribution (Ubuntu 22.04 LTS)

Sandbox Security Measures

- Network isolation using virtual network adapter and firewall rules
- File system snapshot before each execution
- Logging enabled for:
- System calls
- File modifications
- Outbound/inbound network traffic